Student Registration Number

B

Course Code: DMGT204
Course Title: Quantitative Techniques-1

Time Allowed: 3 hours **Max. Marks: 80**

1. *This paper contains 10 questions divided in two parts on __2__ pages.*

2. Part A is compulsory.

3. In Part B (Questions 2 to 10), attempt any 6 questions out of 9. Attempt all parts of the selected question.

4. *The marks assigned to each question are shown at the end of each question in square brackets.*

5. *Answer all questions in serial order*.

6. The student is required to attempt the question paper in English medium only.

PART A

Q1. (a) What is k-step sampling?
(b) Give two examples of negative correlation from the real world.
(c) What is the formula for Paasche's Index?
(d) Write any two characteristics of a normal distribution.
(e) What is the difference between a proposition and hypothesis?
(f)What is stratified random sampling?
(g) Give two examples of positive correlation from the real world.
(h) What is the formula for Laspeyr's Index?
(i) Write any two characteristics of a normal distribution.
(j) What is a null hypothesis? [10x2=20]

PART B

Q2. (a) Seven competitors in a beauty contest were ranked by two judges in the following order given below. Find out if the approach of both judges to beauty is common or not by calculating rank correlation.

Judge A	2	1	4	3	5	7	6
Judge B	1	3	2	4	5	6	7

(b) From a well shuffled pack of 52 cards, a card is drawn at random. Find the probability that is a queen or a king. [10]

Q3 Construct a regression equation of consumption of household(Y) on income of household(X) from the following data. Calculate value of consumption level when household income is 65

Income of household ('000)(X)	50	60	50	60	80	50	80	40	70
Consumption of household ('000)(Y)	30	60	40	50	60	30	70	50	60

[10]

Q4 From the following data, calculate Fishers Index and show that it satisfies the time reversal and factor reversal test. [10]

Commodity	p0	p1	q0	q1
A	2	4	8	6
B	5	6	10	5
C	4	5	14	10
D	2	2	19	13

Q5 Using method of least squares, estimate sales for year 2013 using time series analysis.

Year (X)	2001	2002	2003	2004	2005	2006	2007	2008
Sales (Y)	38	40	65	72	69	60	87	95

[10]

Q6 Write brief notes on the use of statistics in (a) Industry (b) Government departments. [10]

Q7 The following graph shows the monthly expenditure of two families living in Mumbai and Delhi respectively. Show the following data with a subdivided bar graph.

Expenditure Items	**Family A(Mumbai) Total Income Rs. 4000**	**Family B(Delhi) Total Income Rs. 5000**
Food	**1000**	**1200**
Clothing	**800**	**1500**
Housing	**800**	**800**
Heath & Education	**600**	**800**
Fuel & Lighting	**400**	**400**
Miscellaneous	**400**	**300**

[10]

Q8 Calculate Karl Pearson's coefficient of correlation from the following data

Year	2004	2005	2006	2007	2008	2009	2010	2011
Production	100	102	104	107	105	112	103	99

[10]

Q9 Write brief notes on the following methods of sampling: (a) Stratified random sampling (b) cluster sampling. [10]

Q10 Calculate the mode from the following series by method of grouping

Class	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	10	14	15	12	9	10

[10]

Important Formulae

Rank Correlation

$R = 1 - \frac{6\sum D^2}{N^3 - N}$

Karl Pearson's Coefficient of correlation

$\sum xy / \sqrt{\sum x^2 \times \sum y^2}$

Time Series

Method of least Squares;

$Y = a + bX$

$\sum Y = N + b\sum X$

$\sum XY = a\sum X + b\sum X^2$

If $\sum X = 0$, then $a = \frac{\sum Y}{N}$ $b = \frac{\sum XY}{\sum X^2}$

Regression

Regression Equation of Y on X

$Y - Y = r\frac{\sigma y}{\sigma x}(X - X)$

$r\frac{\sigma y}{\sigma x} = \frac{\sum xy}{\sum x^2}$

Mode: 3 Median-2Mean